

Learning the Cross-VAE Inverse Mapping

Latent-Discriminator and Conditional-Flow Methods (Algorithm Pseudocode)

Jayce-Ping

Abstract

Companion to `avoid_inverse_transport_analysis.tex` and the XOPD cross-VAE code (`trainers/xopd`). In the cross-VAE setting with $d_T > d_S$ (teacher VAE has more channels), the forward transport $P : \mathcal{Z}_T \rightarrow \mathcal{Z}_S$ (teacher→student, high→low) is easy and well learned, while the inverse $Q : \mathcal{Z}_S \rightarrow \mathcal{Z}_T$ (student→teacher, low→high) is underdetermined. The current Q is trained by latent MSE, which converges to the *conditional mean* $\mathbb{E}[z_T | z_S, h_S]$; on a curved teacher manifold that mean is *off-manifold*, so the L1 teacher velocity is queried out-of-distribution and training collapses (and the learned Q cannot beat the on-manifold pixel bridge $\mathcal{E}_T(\mathcal{D}_S(z_S))$). This note gives pseudocode for two inverse-learning methods that replace the mean-seeking MSE with *manifold-respecting* objectives: (1) a latent-discriminator (adversarial + cycle-consistency) inverse, and (2) a conditional normalizing-flow inverse trained by maximum likelihood. Both keep the *linear* forward P so the L1 transition-mean pushforward stays exact. We also slot in (3) Cross-Space Bridge as an external Q recipe aimed at Teacher-space VSD/GAN rather than L1 pushforward.

1 Setup and Notation

Frozen VAEs give encoders/decoders $(\mathcal{E}_T, \mathcal{D}_T)$ on the teacher latent space $\mathcal{Z}_T \subseteq \mathbb{R}^{d_T}$ and $(\mathcal{E}_S, \mathcal{D}_S)$ on the student latent space $\mathcal{Z}_S \subseteq \mathbb{R}^{d_S}$, with $d_T > d_S$. The student transformer produces hidden states $h_S = h_S(z_S, t, c)$ (prompt-aware; DeepStack multi-layer). We learn:

- forward $P : \mathcal{Z}_T \rightarrow \mathcal{Z}_S$ — **linear/affine**, closed-form least squares, frozen; keeps the L1 transition-mean pushforward exact ($\mathbb{E}[Pz] = P \mathbb{E}[z]$);
- inverse $Q : \mathcal{Z}_S \rightarrow \mathcal{Z}_T$ — the hard, underdetermined direction; the object both algorithms below learn.

The ideal (on-manifold) inverse target is the pixel bridge $\bar{z}_T(z_S) = \mathcal{E}_T(\mathcal{D}_S(z_S))$; any learned Q must be at least as on-manifold as this. $\text{sg}[\cdot]$ is stop-gradient; noise fraction $\sigma \in [0, 1]$ with $z_\sigma = (1 - \sigma)z + \sigma\varepsilon$, $\varepsilon \sim \mathcal{N}(0, I)$.

2 Method 1: Latent-Discriminator Inverse (CrossNet-style)

Idea (Sendik et al., CrossNet 2020): force $Q(z_S, h_S)$ onto the teacher *latent* manifold with a discriminator on $\mathcal{Z}_T$, and keep P, Q mutually consistent with latent cycle-consistency, instead of regressing to the (off-manifold) mean. Trained offline on a corpus of images; only Q (and, if desired, P) is kept for L1.

3 Method 2: Conditional-Flow Inverse (cINN-style, the fundamental fix)

Idea (Ardizzone/Rombach et al., conditional INN): model the full conditional distribution $p(z_T | z_S, h_S)$ with a conditional normalizing flow F trained by maximum likelihood, instead of a

Algorithm 1 Latent-Discriminator Inverse — offline training

Require: frozen $\mathcal{E}_T, \mathcal{D}_T, \mathcal{E}_S$, student transformer; linear P (pre-fit); inverse Q_ψ ; latent discriminator $\text{Disc}_\phi : \mathcal{Z}_T \rightarrow \mathbb{R}$; weights $\lambda_{\text{cyc}}, \lambda_{\text{rec}}$; max noise σ_{max} .

```
1: for each minibatch of images  $x$  do
2:    $z_T \leftarrow \mathcal{E}_T(x)$  ▷ clean teacher latent (real sample on  $\mathcal{Z}_T$ )
3:    $z_S \leftarrow \mathcal{E}_S(x); \quad \sigma \sim U(0, \sigma_{\text{max}}); \quad z_S^\sigma \leftarrow (1 - \sigma)z_S + \sigma\varepsilon$  ▷ cross- $\sigma$  (on-policy-like)
4:    $h_S \leftarrow \text{STUDENTHIDDEN}(z_S^\sigma, c, \sigma)$  ▷ DeepStack maps, no grad
5:    $\hat{z}_T \leftarrow Q_\psi(z_S^\sigma, h_S)$ 
   - update  $Q_\psi$  (generator) -
6:    $\mathcal{L}_{\text{adv}} \leftarrow (\text{Disc}_\phi(\hat{z}_T) - 1)^2$  ▷ LS-GAN: push  $\hat{z}_T$  onto the teacher latent manifold
7:    $\mathcal{L}_{\text{cyc}} \leftarrow \|P(\hat{z}_T) - z_S\|^2 + \|Q_\psi(P(z_T), h_S) - z_T\|^2$  ▷  $P \circ Q, Q \circ P$  bijection
8:    $\mathcal{L}_{\text{rec}} \leftarrow \|\mathcal{D}_T(\hat{z}_T) - x\|_1$  ▷ weak on-manifold anchor
9:    $\psi \leftarrow \psi - \eta \nabla_\psi (\mathcal{L}_{\text{adv}} + \lambda_{\text{cyc}} \mathcal{L}_{\text{cyc}} + \lambda_{\text{rec}} \mathcal{L}_{\text{rec}})$ 
   - update  $\text{Disc}_\phi$  -
10:   $\mathcal{L}_D \leftarrow (\text{Disc}_\phi(z_T) - 1)^2 + (\text{Disc}_\phi(\text{sg}[\hat{z}_T]))^2$ 
11:   $\phi \leftarrow \phi - \eta_D \nabla_\phi \mathcal{L}_D$ 
12: end for
13: return  $Q_\psi$  ▷ used as the L1 inverse; query point is deterministic  $Q_\psi(z_S, h_S)$ 
```

point estimate. Sampling F produces *on-manifold* teacher latents by construction; the mean-seeking failure disappears. We model the *residual* around the linear pseudo-inverse base $b(z_S) = P^+(z_S - \beta)$ so a zero-initialised flow starts exactly at the linear baseline (do-no-harm).

Algorithm 2 Conditional-Flow Inverse — offline training (NLL)

Require: frozen $\mathcal{E}_T, \mathcal{D}_T, \mathcal{E}_S$, student transformer; linear P (pre-fit), base $b(\cdot) = P^+(\cdot - \beta)$; conditional flow F_ψ (base dist $\mathcal{N}(0, I)$); weight λ_{px} ; max noise σ_{max} .

```
1: for each minibatch of images  $x$  do
2:    $z_T \leftarrow \mathcal{E}_T(x); \quad z_S \leftarrow \mathcal{E}_S(x)$ 
3:    $\sigma \sim U(0, \sigma_{\text{max}}); \quad z_S^\sigma \leftarrow (1 - \sigma)z_S + \sigma\varepsilon; \quad h_S \leftarrow \text{STUDENTHIDDEN}(z_S^\sigma, c, \sigma)$ 
4:    $c_{\text{cond}} \leftarrow \text{FUSE}(z_S^\sigma, h_S)$  ▷ conditioning tensor
5:    $r \leftarrow z_T - b(z_S^\sigma)$  ▷ residual to model
6:    $(v, \text{logdet}) \leftarrow F_\psi.\text{FORWARD}(r \mid c_{\text{cond}})$  ▷ normalize to base space
7:    $\mathcal{L}_{\text{nll}} \leftarrow \frac{1}{2} \|v\|^2 - \text{logdet}$  ▷  $-\log p_\psi(z_T \mid z_S, h_S)$  up to const
8:    $\mathcal{L}_{\text{px}} \leftarrow \|\mathcal{D}_T(b(z_S^\sigma) + F_\psi.\text{INVERSE}(\mathbf{0} \mid c_{\text{cond}})) - x\|_1$  ▷ optional mode ( $v=0$ ) anchor
9:    $\psi \leftarrow \psi - \eta \nabla_\psi (\mathcal{L}_{\text{nll}} + \lambda_{\text{px}} \mathcal{L}_{\text{px}})$ 
10: end for
11: return  $F_\psi$ 
```

4 Method 3: Cross-Space Bridge

External recipe for the same hard inverse $Q : \mathcal{Z}_S \rightarrow \mathcal{Z}_T$, designed for **Teacher-space** VSD/GAN (DMD2-style one-step distillation), not for XOPD L1 pushforward. Full write-up: docs/xopd/cross_vae/cross_space_distillation_bridge.tex.

Contrast with Methods 1–2. Bridge also targets on-manifold $\hat{z}_T$, but (i) uses a *frozen Student-decoder prefix* instead of student DiT hidden states, (ii) uses Teacher *attention* reverse KL instead of a latent discriminator or cINN likelihood, and (iii) is paired with distribution matching *in* $\mathcal{Z}_T$, so it does not need to keep a linear forward P for L1. To reuse Bridge inside

Algorithm 3 Conditional-Flow Inverse — L1 teacher query (inside `transition_mean_to_student`)

Require: student on-policy state z_S (+ its h_S); trained F_ψ , base b , linear P ; teacher transition operator `TEACHERNEXT`($\cdot$); `query_mode` $\in \{\text{mode}, \text{sample}, \text{mean_k}\}$.

```

1:  $c_{\text{cond}} \leftarrow \text{FUSE}(z_S, h_S)$ 
2: if query_mode = mode then
3:    $v \leftarrow \mathbf{0}$ 
4: else if query_mode = sample then
5:    $v \sim \mathcal{N}(\mathbf{0}, I)$ 
6: else ▷ mean_k
7:    $v \leftarrow$  (average over  $K$  draws handled below)
8: end if
9:  $\hat{z}_T \leftarrow b(z_S) + F_\psi.\text{INVERSE}(v \mid c_{\text{cond}})$  ▷ on-manifold teacher latent (mean_k: average  $\hat{z}_T$  over  $K$  draws)
10:  $\mu_T \leftarrow \text{TEACHERNEXT}(\hat{z}_T)$  ▷ teacher next-step mean in  $\mathcal{Z}_T$ 
11: return  $z_S + (P(\mu_T) - P(\hat{z}_T))$  ▷ displacement anchoring: cancels  $P$ 's self-recon bias (velocity matching)

```

Algorithm 4 Conditional affine-coupling block (one step of F_ψ)

Require: input $u \in \mathbb{R}^{C \times H \times W}$, condition c_{cond} ; split index $C/2$; zero-initialised head net g .

```

1:  $u \leftarrow \text{ACTNORM}(u)$ ; accumulate logdet +=  $HW \sum_c \log |s_c^{\text{an}}|$ 
2:  $u \leftarrow \text{PERMUTE}(u)$  ▷ fixed channel flip (mixes halves across blocks)
3:  $(u_a, u_b) \leftarrow \text{SPLIT}(u)$ 
4:  $(\log s, \tau) \leftarrow g([u_b, c_{\text{cond}}])$  ▷  $g$  last layer zero-init  $\Rightarrow$  identity at start
5:  $u'_a \leftarrow u_a \odot \exp(\log s) + \tau$ ; logdet +=  $\sum \log s$ 
6: return  $[u'_a, u_b]$  ▷ inverse: undo  $\tau, \exp$ , then inverse permute/actnorm

```

XOPD L1, still need P (or Route B pixel adjoint); the Attention Fidelity term alone is a drop-in regularizer for any Q .

5 Notes

- **Linear P kept.** Methods 1–2 only change the inverse Q ; the forward P stays a frozen closed-form affine, so the L1 transition-mean pushforward remains exact and analytically consistent (`docs/xopd/cross_vae/xopd_vae_space_align.tex`, Prop. affine). Method 3 (Bridge) typically *does not* use P during distillation.
- **Do-no-harm init.** Modelling the residual $r = z_T - b(z_S)$ with a zero-initialised flow (Alg. 4) means at step 0, $F.\text{INVERSE}(\mathbf{0} \mid c) = \mathbf{0}$ and $\hat{z}_T = b(z_S) = P^+ z_S$ — exactly the current linear pseudo-inverse baseline; the flow can only improve on-manifoldness.
- **Why this fixes the collapse.** A deterministic MSE Q returns the conditional mean (off-manifold). Method 1's discriminator and Method 2's likelihood both reward *distribution membership*, so $\hat{z}_T$ lands on $\mathcal{Z}_T$ and the teacher is queried in-distribution; the L1 guidance becomes meaningful. Bridge's Attention Fidelity plays an analogous role for Teacher-space VSD.
- **L1 mean matching.** D_k matches transition means; default `query_mode` = `mode` ($v=0$, deterministic, on-manifold central point). `sample` injects stochastic-but-on-manifold teacher opinions; `mean_k` averages K samples to approximate $\mathbb{E}[z_T \mid \cdot]$ at $K \times$ cost.

Algorithm 5 Cross-Space Bridge — offline training (summary)

Require: frozen $\mathcal{E}_T, \mathcal{E}_S$, Student decoder prefix $\mathcal{D}_S^{(n)}$ (frozen; $n=1$ expands grid to Teacher resolution); projector g_ϕ (e.g. SwinIR); Teacher denoiser for attention maps; weights α, β (default 1); temperature τ .

- 1: **for** each minibatch of images x **do**
 - 2: $z_T \leftarrow \mathcal{E}_T(x); \quad z_S \leftarrow \mathcal{E}_S(x)$
 - 3: $f_{\text{prior}} \leftarrow \mathcal{D}_S^{(n)}(z_S)$ $\triangleright$ spatial prior: $h \times w \rightarrow H \times W$, no trainable params
 - 4: $\hat{z}_T \leftarrow g_\phi(f_{\text{prior}})$
 - 5: $\mathcal{L}_{\text{rec}} \leftarrow \|z_T - \hat{z}_T\|_1$
 - 6: $\mathcal{L}_{\text{attn}} \leftarrow \sum_l \tau^2 \text{KL}(P^l(\hat{z}_T; t, c) \| P^l(z_T; t, c))$ $\triangleright$ reverse KL on Teacher self-attention; $t=1$ in paper
 - 7: $\phi \leftarrow \phi - \eta \nabla_\phi(\alpha \mathcal{L}_{\text{rec}} + \beta \mathcal{L}_{\text{attn}})$
 - 8: **end for**
 - 9: **return** $\mathcal{B}_\phi = g_\phi \circ \mathcal{D}_S^{(n)}$ $\triangleright$ freeze; map Student latents into Teacher space for VSD/GAN
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